

CONCEPT QUESTIONS – CH : 3

Steady Heat Conduction in Plane Walls

3-1C

Consider heat conduction through a wall of thickness L and area A . Under what conditions will the temperature distribution in the wall be a straight line?

The temperature distribution in the wall will be a straight line under steady-state, one-dimensional, no heat generation, and thermal conductivity is constant.

3-2C

Consider heat conduction through a plane wall. Does the energy content of the wall change during steady heat conduction? How about during transient conduction? Explain.

Steady-State: No, since the wall's energy storage does not change with time.

$$\frac{d\dot{E}_{st}}{dt} = 0 \rightarrow \dot{E}_{in} = \dot{E}_{out}$$

Transient: Yes, since the wall energy storage does change with time.

$$\frac{d\dot{E}_{st}}{dt} = \dot{E}_{in} - \dot{E}_{out}$$

3-3C

What does the thermal resistance of a medium represent?

The thermal resistance of a medium represents the resistance to heat flow or heat flux.

3-4C

Can we define the convection resistance for a unit surface area as the inverse of the convection heat transfer coefficient?

Yes, the convection resistance is equal to the inverse of the convection heat transfer coefficient.

$$R_{conv} = \frac{1}{hA} \text{ for unit surface area } A=1 \rightarrow R_{conv} = \frac{1}{h}.$$

3-5C

Consider steady heat transfer through the wall of a room in winter. The convection heat transfer coefficient at the outer surface of the wall is three times that of the inner surface as a result of winds. On which surface of the wall will the temperature be closer to the surrounding air temperature? Explain.

The outer surface will be closer to the surrounding air temperature. The reason is that a larger convection heat transfer coefficient means a small convection resistance resulting in a smaller temperature drop.

3-6C

How is the combined heat transfer coefficient defined? What convenience does it offer in heat transfer calculations?

The combined heat transfer coefficient is the sum of convection and radiation at a surface. This combines both heat transfer modes and all complications with radiation are avoided.

$$h_{conv} + h_{rad} = h_{combined}$$

3-7C

Why are the convection and the radiation resistances at a surface in parallel instead of being in series?

Radiation and convection are in parallel because both heat transfers occur simultaneously between surface and surrounding. Furthermore, heat leaves/enters the surface at the same time which makes their thermal resistance act in parallel.

3-8C

Consider steady one-dimensional heat transfer through a plane wall exposed to convection from both sides to environments at known temperatures T_1 and T_2 with known heat transfer coefficients h_1 and h_2 . Once the rate of heat transfer Q has been evaluated, explain how you would determine the temperature of each surface.

Once heat transfer (Q) is known apply convection heat transfer (Newton's 2nd Law of Cooling) equations to find $T_{surface}$ by rearranging the equation.

3-9C

Someone comments that a microwave oven can be viewed as a conventional oven with zero convection resistance at the surface of the food. Is this an accurate statement?

No, since a microwave uses electromagnetic waves that absorb into the food to generate heat. Therefore, heat transfer in a microwave oven does not have zero convection resistance.

3-10C

Consider two cold canned drinks, one wrapped in a blanket and the other placed on a table in the same room. Which drink will warm up faster?

The drink placed on the table will warm up faster. The blanket acts as an insulator which increases thermal resistance between the drink and the surrounding.

3-11C

The bottom of a pan is made of a 4-mm-thick aluminum layer. In order to increase the rate of heat transfer through the bottom of the pan, someone proposes a design that consists of a 3-mm-thick copper layer sandwiched between two 2-mm-thick aluminum layers. Will the new design conduct heat better? Explain. Assume perfect contact between the layers.

No, since the new design adds more resistance because of the additional layers which increases the conduction resistance of the pan. While copper has higher thermal conductivity the added layers result in lower heat transfer rate.

3-12C

Consider a surface of area A at which the convection and radiation heat transfer coefficients are h_{conv} and h_{rad} , respectively. Explain how you would determine (a) the single equivalent heat transfer coefficient, and (b) the equivalent thermal resistance. Assume the medium and the surrounding surfaces are at the same temperature.

(a) Since convection and radiation act simultaneously use $h_{conv} + h_{rad} = h_{combined}$.

(b) The equivalent resistance is $R_{combined} = \frac{1}{h_{combined}A}$

3-13C

How does the thermal resistance network associated with a single-layer plane wall differ from the one associated with a five-layer composite wall?

Single Layer: One conduction for the wall.

Five Layer: 5 conduction resistances in series along the heat-flow path.

3-14C

Consider steady one-dimensional heat transfer through a multilayer medium. If the rate of heat transfer $\dot{Q}$ is known, explain how you would determine the temperature drop across each layer.

Since $\dot{Q}$ is the same across each layer in steady one-dimensional heat transfer. The temperature drop across the layer can be determined by:

$$\dot{Q}R = \Delta T$$

3-15C

Consider a window glass consisting of two 4-mm-thick glass sheets pressed tightly against each other. Compare the heat transfer rate through this window with that of one consisting of a single 8-mm-thick glass sheet under identical conditions.

Since the total thickness of the two windows is the same and for the two 4-mm glass sheets assumed perfect contact, they have the same conduction resistance and identical boundary conditions for heat transfer rates.

$$R_{cond} = \frac{L}{kA}$$

Thermal Contact Resistance

3-43C

What is thermal contact resistance? How is it related to thermal contact conductance?

Thermal contact resistance is the resistance to heat flow at two surfaces in contact. Thermal contact resistance is the inverse of thermal contact conductance.

3-44C

Will the thermal contact resistance be greater for smooth or rough plain surfaces?

Thermal contact resistance will be greater for rough plain surfaces since rough plain surfaces create air gaps which increases resistance to heat flow.

3-45C

Explain how the thermal contact resistance can be minimized.

Thermal contact resistance can be minimized by increasing contact area between surfaces to decrease the number of air gaps. This can be achieved by smoothing the surfaces, increasing contact pressure, and using soft materials to eliminate air gaps.

3-46C

A wall consists of two layers of insulation pressed against each other. Do we need to be concerned about the thermal contact resistance at the interface in a heat transfer analysis, or can we just ignore it?

The thermal contact resistance can be ignored since the material is soft creating good contact when pressed together creating very low thermal contact resistance.

3-47C

A plate consists of two thin metal layers pressed against each other. Do we need to be concerned about the thermal contact resistance at the interface in a heat transfer analysis, or can we just ignore it?

Yes, since the metal surface is normally rough and rigid, creating limited number of contact points which will significantly increase the thermal contact resistance.

3-48C

Consider two surfaces pressed against each other. Now the air at the interface is evacuated. Will the thermal contact resistance at the interface increase or decrease as a result?

The thermal contact resistance will increase at the interface since that is the only area where heat transfer can occur due to the removal of air between the two surfaces.

Generalized Thermal Resistance Networks

3-58C

What are the two approaches used in the development of the thermal resistance network for two-dimensional problems?

Parallel heat flow path: each path is treated as a 1-D resistance then combined as parallel

Isothermal planes normal to heat flow: Temperature is assumed uniform allowing for reduction to equivalent resistance network.

The first approach tends to overpredict thermal resistance, while the second tends to underpredict thermal resistance.

3-59

The thermal resistance networks can also be used approximately for multidimensional problems. For what kind of multidimensional problems will the thermal resistance approach give adequate results?

Thermal resistance will give adequate results when the heat flow path can be approximated as one-dimensional. This is the case when temperature variation is small and the geometry / boundary conditions can be represented as parallel or series in one-dimensional.

3-60

When plotting the thermal resistance network associated with a heat transfer problem, explain when two resistances are in series and when they are in parallel.

Two resistances are in series when heat flows through both resistances in the same path. This is when the heat flow passes through each resistance sequentially

Two resistances are in parallel when heat can flow through more than one path simultaneously. This results in same temperature difference across each individual path.

Heat Conduction in Cylinders & Spheres

3-73C

What is an infinitely long cylinder? When is it proper to treat an actual cylinder as being infinitely long, and when is it not?

Infinitely long cylinders are when ends are negligible and heat transfer only occurs in radial direction. It is proper to treat a cylinder as infinitely long when the length is a lot longer than its radius which allows axial heat transfer to be neglected. It is not proper to use infinitely long cylinder when it is short due to the significance of axial and radial heat transfer.

3-74C

Can the thermal resistance concept be used for a solid cylinder or sphere in steady operation? Explain.

Yes, thermal resistance concepts for a solid cylinder and sphere can be used in steady state. The heat conduction equations for both cylindrical and spherical object can be solved in the form of equivalent thermal resistance.

$$R_{cyl} = \frac{\ln(r_2/r_1)}{2\pi Lk}$$

$$R_{sphere} = \frac{r_2 - r_1}{4\pi k r_1 r_2}$$

3-75C

Consider a short cylinder whose top and bottom surfaces are insulated. The cylinder is initially at a uniform temperature T_i and is subjected to convection from its side surface to a medium at temperature T , with a heat transfer coefficient of h . Is the heat transfer in this short cylinder one- or two-dimensional? Explain.

The heat transfer in the short cylinder is one-dimensional since there is no heat transfer in the axial direction due to both top and bottom of the cylinder being insulated. This causes no heat flow in the axial direction due to adiabatic relationship and heat flow primarily in radial direction making it a one-dimensional problem.

Critical Radius of Insulation

3-96C

What is the critical radius of insulation? How is it defined for a cylindrical layer?

The critical radius of insulation is the outer radius of insulation at which the rate of heat transfer is maxed. For a cylinder layer, the critical radius is $r_{cr,cylidner} = k/h$ where k is thermal conductivity of the insulation and h is the external convection heat coefficient.

3-97C

Consider an insulated pipe exposed to the atmosphere. Will the critical radius of insulation be greater on calm days or on windy days? Why?

The critical radius will be greater on calm days since the convection heat transfer coefficient h is small which increases the critical radius of an insulated pipe.

3-98C

A pipe is insulated to reduce the heat loss from it. However, measurements indicate that the rate of heat loss has increased instead of decreasing. Can the measurements be right?

Yes, since adding insulation will increase the heat transfer rate due to the increase in surface area which reduces convection resistance.

3-99C

Consider a pipe at a constant temperature whose radius is greater than the critical radius of insulation. Someone claims that the rate of heat loss from the pipe has increased when some insulation is added to the pipe. Is this claim valid?

No, since the pipe radius is greater than the critical radius of insulation, adding insulation will increase thermal resistance which will reduce heat loss.

3-100C

A pipe is insulated such that the outer radius of the insulation is less than the critical radius. Now the insulation is taken off. Will the rate of heat transfer from the pipe increase or decrease for the same pipe surface temperature?

The rate of heat transfer will decrease since the removal of insulation will increase convection resistance by reducing the outer surface area. Below the critical radius, removing insulation will increase convection resistance and therefore decrease heat transfer rate.

Heat Transfer from Finned Surfaces

3-105C

Hot air is to be cooled as it is forced to flow through tubes exposed to atmospheric air. Fins are to be added in order to enhance heat transfer. Would you recommend attaching the fins inside or outside the tubes? Why? When would you recommend attaching fins both inside and outside the tubes?

Fins should be attached to the outside tubes since atmospheric has a low convection coefficient and it is where convection heat transfer coefficient is the smallest. The only time where fins should be on both the inside and outside is when convection heat transfer coefficients are both low and similar in magnitude. Typical when there are two gas fluids.

3-106C

What is the reason for the widespread use of fins on surfaces?

Fins are used to increase the effective surface area for convection, which generates more heat transfer. This is important when the heat transfer coefficient is small.

3-107C

What is the difference between the fin effectiveness and the fin efficiency?

Fin effectiveness is the ratio of heat transfer rate from a fin to the same surface area without the fin.

Fin efficiency is the ratio of actual heat transfer rate from a fin to the max possible heat transfer rate from a fin.

3-108C

The fins attached to a surface are determined to have an effectiveness of 0.9. Do you think the rate of heat transfer from the surface has increased or decreased as a result of the addition of these fins?

The rate of heat transfer would decrease since the effectiveness value is less than 1. This means that the fin acts more like insulation which reduces the overall heat transfer rate.

3-109C

Explain how the fins enhance heat transfer from the surface. Also, explain how the addition of fins may decrease heat transfer from the surface.

Fins enhance heat transfer by increasing the surface area for convection, causing the overall thermal resistance to decrease and heat transfer rate to increase.

Fins decrease heat transfer by having a fin effectiveness less than 1. This causes the fins to act as insulation and increase conduction resistance causing heat transfer rate to decrease.

3-110C

How does the overall effectiveness of a finned surface differ from the effectiveness of a single fin?

The overall effectiveness of a finned surface compares the total heat transfer to the entire finned surface including the base to the same surface without fins. The effectiveness of a single fin only compares the heat transfer rate of one fin to the heat transfer rate of the same base without the fin.

3-111C

Hot water is to be cooled as it flows through tubes exposed to atmospheric air. Fins are to be attached in order to enhance heat transfer. Would you recommend attaching the fins inside or outside the tubes? Why?

Fins should be attached to the outside of the tubes since atmospheric air has a low convection coefficient compared to hot water. Fins are most effective when they are placed on the side with larger thermal resistance, which means it is on the side with the lowest h .

3-112C

Consider two finned surfaces that are identical except that the fins on the first surface are formed by casting or extrusion, whereas they are attached to the second surface afterwards by welding or tight fitting. For which case do you think the fins will provide greater enhancement in heat transfer? Explain.

The casting or extruded fins would be better to enhance the heat transfer rate as it will have a lower thermal resistance compared to the welded or tight-fitting fins. The extruded or casting fins have better heat conduction from the surface as they won't have thermal contact resistance compared to the welded or tight-fitting fins.

3-113C

The heat transfer surface area of a fin is equal to the sum of all surfaces of the fin exposed to the surrounding medium, including the surface area of the fin tip. Under what conditions can we neglect heat transfer from the fin tip?

Heat transfer of a fin can be neglected when the fin is long and thin. This is because at the fin tip the temperature will be close to the surrounding fluid temperature, which results in a very low temperature difference.

3-114C

Does the (a) efficiency and (b) effectiveness of a fin increase or decrease as the fin length is increased?

- (a) The fin efficiency decreases as the fin length increases since the overall temperature difference along the fin between two points close together becomes smaller.
- (b) The fin effectiveness increases as the fin length increases since the surface area for convection will continue to increase, which will increase the total heat transfer rate.

3-115C

Two pin fins are identical, except that the diameter of one of them is twice the diameter of the other. For which fin is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

- (a) The fin effectiveness is higher for the smaller diameter pin fin. A smaller diameter fin has a smaller base area, which causes the heat transfer from the fin to be large compared to the same base area without the fin.
- (b) The fin efficiency is higher for the larger diameter pin fin. A larger diameter will have a larger cross-sectional area which will reduce the overall temperature drop along the fin.

3-116C

Two plate fins of constant rectangular cross section are identical, except that the thickness of one of them is twice the thickness of the other. For which fin is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

- (a) The fin effectiveness is higher for the smaller thinner fin. A smaller thickness fin has a smaller base area, which causes the heat transfer from the fin to be large compared to the same base area without the fin.
- (b) The fin efficiency is higher for the thicker fin. A larger thickness will have a larger cross-sectional area which will reduce the overall temperature drop along the fin.

3-117C

Two finned surfaces are identical, except that the convection heat transfer coefficient of one of them is twice that of the other. For which finned surface is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

- (a) The fin effectiveness is higher for the smaller convection heat transfer coefficient. A small convection coefficient transfers very little heat on unfinned surfaces, therefore, adding fins will increase the surface area and increase heat transfer.
- (b) The fin efficiency is higher for the larger convection heat transfer coefficient. A higher convection coefficient allows for more heat to be removed from a fin surface which allows the fin to operate more effectively, resulting in higher efficiency.